National University of Computer and Emerging Sciences
Karachi Campus

Digital Logic Design (EE1005)

Date: February 24th 2026

Course Instructors:

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Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 24

Total Questions: 3

Roll No ______ Section ______ Student Signature ______

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Attempt all the questions in the given sequence.

CLO # 1: Identify and explain fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded system, basic components of combinational and sequence circuits.

Number System Conversions, BCD, and Signed Number Representations [4 x 2 = 08 Marks]

Time: 20 Minutes

Q1: (a) Convert the following numbers to the specified system. Show all steps clearly.

(i) $(255.12)_{10}$ to binary 11111111.0001 (2 to 3 decimal places answer can also be considered as correct)

(ii) $((01101011.0011)_2$ to hexadecimal.

$(6B.3)_{16}$

(b) Represent −37 in:

(i) 8-bit Sign-Magnitude +37= 00100101 → -37= 10100101

(ii) 1's Complement 00100101→11011010

(c) Determine how many bits change when transitioning from: $10101101_2 \rightarrow 10101110_2$ in:

(i) Binary representation: 10101101→10101110 **Two bits change** (last two bits).

(ii) Gray code representation: Gray equivalent: $10101101_2 \rightarrow 11111011$

$10101110_2 \rightarrow 11111001$ Only **one bit changes** in Gray code.

(d) Perform the following operations. Show all steps clearly.

(i) Perform the subtraction using 2's complement method and indicate whether overflow occurs: $10101101_2 - 01110110_2$

2's complement of subtrahend 01110110=10001010

Add to Minuend: 10101101 + 10001010 **= 1 00110111**

We get a carry out = 1 Discard the carry (since 8-bit system):

Result=00110111

(iii) Perform BCD correction wherever required and give the final answer in valid BCD form:

```
     1
   0110 0100
 + 0011 0111
-------------
0001 0000 0001
```

0001 0000 0001

0100+0111 = 1011+ 0110 = 1 0001

0110 + 1 + 0011 = 1010 + 0110 = 1 0000

CLO # 2: Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronics circuits, including Boolean Algebra and Multi-variable Karnaugh map methods.

***Design and analysis of digital electronics circuits*: [4 x 4 = 16 Marks]**

Time: 40 Minutes

Q2: (a) A factory monitoring system has three inputs:

A = Temperature high (1 = Yes) B = Pressure high (1 = Yes)

C = Gas detected (1 = Yes)

The alarm F should turn ON when:

- Temperature AND Pressure are high
- Temperature AND Gas are detected
- Pressure AND Gas are detected

(i) Write the Boolean expression for the given system. **F = AB + AC + BC**

(ii) Construct the truth table corresponding to the Boolean expression.

A	B	C	AB	AC	BC	F
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	0	0	0	0
1	0	1	0	1	0	1
1	1	0	1	0	0	1
1	1	1	1	1	1	1

(b) In a digital communication system, data is transmitted in 8-bit format along with a parity bit for error detection. The system currently uses even parity. The transmitted 8-bit data is: **10110110**

(i) Determine whether the above data satisfies even parity. Show your working.

Count the Number of 1's : 1 0 1 1 0 1 1 0 Number of 1's = **5**

Check Even Parity Condition: Even parity requires **total number of 1's to be even.**

Here, number of 1's = **5 (odd)**.

Since the total number of 1's is **odd**, The data does NOT satisfy even parity.

(ii) Draw a logic circuit to generate the parity bit for the given 8-bit input. Also, explain how parity checking detects a single-bit error at the receiver side. [Hint: use XOR gates]

Let the inputs be: A,B,C,D,E,F,G,H

To generate an **even parity bit (P)**: $P=A\oplus B\oplus C\oplus D\oplus E\oplus F\oplus G\oplus H$ (**Draw a circuit from it**)

($\oplus$ = XOR operation), Therefore:

- If input has **odd 1's → P = 1**
- If input has **even 1's → P = 0**

At the receiver:

All received bits (including parity bit) are XORed together.

- If result = 0 (for even parity system) → No error
- If result = 1 → Error detected

Parity can detect:

- Any **single-bit error**

(c) Simplify the expression **(AB + AC)' + A'B'C** using Boolean algebra techniques, clearly citing the relevant Rules / Laws, and draw the simplified logic diagram.

Solution: **= (AB)' . (AC)' + A' B' C** Apply De Morgan
= (A'+B'). (A'+C') +A'B'C Apply De Morgan
= A'A'+A'C'+A'B'+B'C'+A'B'C
= A' +A'C'+A'B'(1+C)+B'C'
= A'(1+C')+A'B'+B'C'
= A'+A'B'+B'C'
= A'(1+B')+B'C'
= A' + B'C'

(d) Apply DeMorgan's Theorems to the following expression: **F(A,B,C,D) =(A+B'C)'+A'CD** and simplify it to **F=A'B + A'C' + A'D** using either Boolean algebra (clearly citing the relevant Rules / Laws) or a Karnaugh Map (K-map)

(A+B'C)'=A'(B'C)' **Apply DeMorgan**
F=A'(B+C')+A'CD **Apply DeMorgan**

F=A'B+A'C'+A'CD **Apply Distributive Law**

F=A'(B+C'+CD)

Simplify C'+CD

Use the identity:

X+YZ=(X+Y)(X+Z)
Factor using absorption idea:
C'+CD=(C'+C)(C'+D)
Since: C'+C=1
So: C'+CD=1(C'+D)= C'+D
F=A'(B+C'+D)
F=A'B+A'C'+A'D **Apply Distribute Law**